GUIDE 2

Getting Test Data for your Magic MA Expert Advisor

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Introduction and overview.

The objectives of Guide 2 are to help you:-

- Create a separate test facility for testing all your Expert Advisors;
- Download a considerable amount of historical test data.

Background.

All experienced Forex traders know that certain Forex trading techniques stop working efficiently at some stage. Does this mean that the Expert Advisor or technique is bad? No, it just means that the market has changed to be less ideal for your current settings or technique. With experience you will realise that there is no such thing as a bad technique or Expert Advisor. There are only good techniques or Expert Advisors used in inappropriate market conditions or with poor settings. The same will apply to the Magic Moving Average Expert Advisor. Its efficiency will change over time.

When purchasing a new Expert Advisor there are many variables that will make it work effectively. The main ones are currency crosses and time frames. Given that there may be 12 potential currency crosses and 9 possible time frames you already have 108 possible variations that need testing and optimising to get more favourable settings for any Expert Advisor. Often there are other variables such as day of the week or time of the day that also play a role. All of this even before we get to the variations in actual Expert Advisor settings such as stops, targets, indicator periods, lot sizing etc.

Testing and optimisation.

It is important to do some optimisation and testing before trading any Expert Advisor to have a good feel of how these variables affect each other and your possible outcomes. This guide gives you instructions on how to get data to test your Magic Moving Average Expert Advisor.

When doing the testing and optimising of your Magic Moving Average Expert Advisor, you take **ownership** and **responsibility** for the ongoing results achieved by the Expert Advisor. The test results should have a positive impact on your **belief system** and trading **confidence**. These are important characteristics of successful Forex traders.

The guides that follow will then give you some assistance on how to optimise and test your Magic Moving Average Expert Advisor.

This guide will cover step 2 of the optimisation and test process -

1. Understanding the trading technique and installing your Expert Advisor;
2. Ensuring you have reasonably good test data;
3. Setting your Strategy Tester up and Testing your Expert Advisor;
4. Optimising your Expert Advisor.

Please remember to read the Questions and Answers at the end of this Guide.

We use Alpari UK to show examples in this guide. You can use any MetaTrader 4 Broker to do the process below. When using non Alpari Brokers the steps detailed in this and other guides may vary. Focus on what we are trying to achieve and you should be able to work around any difficulties. We use UK English spelling throughout in our Guides.

When we open a MetaTrader 4 demo account it comes with some basic historical data mainly generated by the broker. This is normally good enough to get you trading and allows for some limited back testing on the charts.

What we want is as much data as we can get and data of reasonable quality. Such data is supplied by MetaQuotes.net, the main MetaTrader website.

The other thing that we want to do is have a test environment separate from our normal MetaTrader trading platform. There are many reasons for this but the main one is that when we download the test data from MetaQuotes we want that data to be pure. If the data is downloaded into a live trading environment the data will become mixed with updates from your own broker. So you could end up with test data from your own broker and MetaQuotes in the same database.

The collecting of test data process we follow is:

1. Open a new MetaTrader test platform on your computer;
2. Make sure that the maximum data can be downloaded and used;
3. Delete all the current data;
4. Load the new data;
5. Protect the test data.

Note to beginners: You may have to follow this process parrot fashion (step by step), not fully understanding what you are doing. Don't worry about it at this stage. It will all start making sense once you have completed all the Guides and you are using your Expert Advisor.

So let's begin.

IMPORTANT

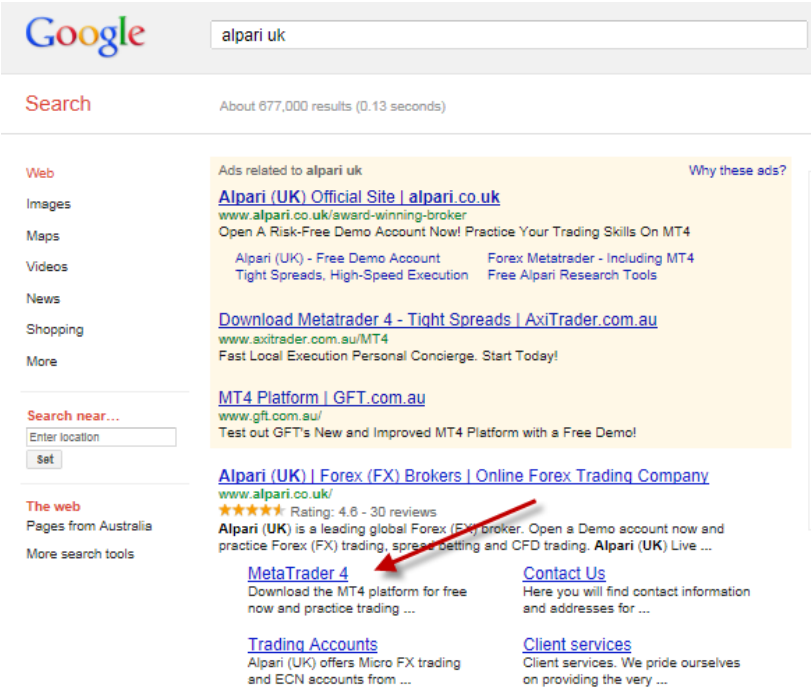
Try to do all data downloads when the Forex Market is open and volatile – The UK or US markets are open in their mornings. This is because the spreads used for testing are sometimes the spreads existing at the time of download.

Do not do any live (connected to the internet) downloading, testing or optimisation over weekends. Brokers tend to increase their spreads considerably over weekends.

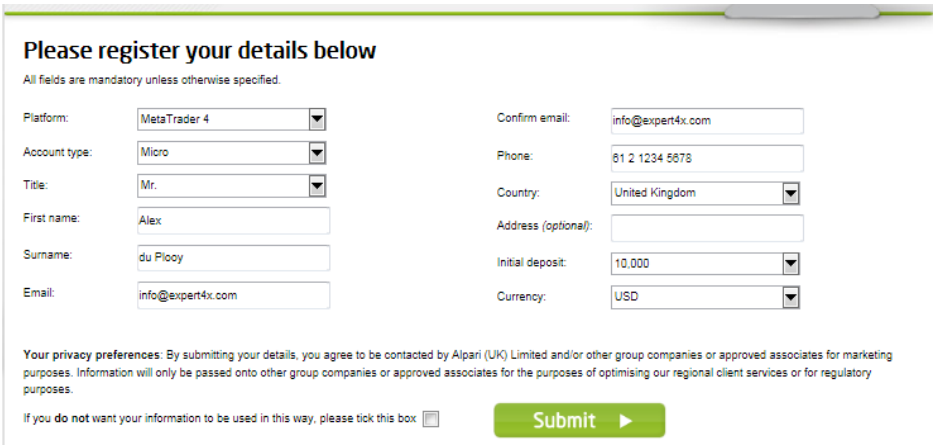
1. Open a new MetaTrader test platform on your computer.

If you already have a MetaTrader platform installed on your computer don't worry. You can have as many as you want. Using Alpari UK as an example (you can use any MetaTrader 4 broker) the first step is to open a new demo account. The steps that follow this Guide may vary if you are using a non Alpari UK broker but you should be able to achieve the same objectives.

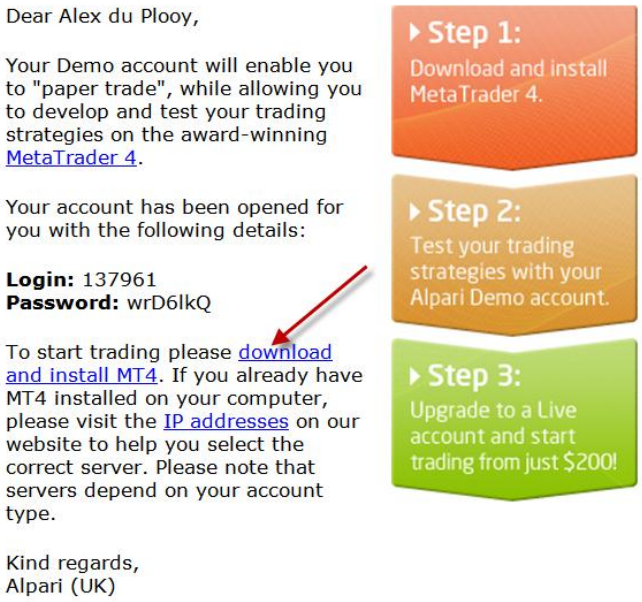
This you do by going to your broker's website. We Googled "Alpari UK" and clicked on the "MetaTrader 4" option. Follow the option "Try MT4 demo".



Complete your details – see the guide below and submit them. It is essential you use a valid email address. You should also select a Micro Account type.



You will receive an email like the one below. Click on the link to “download and install MT4”.



Dear Alex du Plooy,

Your Demo account will enable you to "paper trade", while allowing you to develop and test your trading strategies on the award-winning MetaTrader 4.

Your account has been opened for you with the following details:

Login: 137961
Password: wrD6lkQ

To start trading please download and install MT4. If you already have MT4 installed on your computer, please visit the IP addresses on our website to help you select the correct server. Please note that servers depend on your account type.

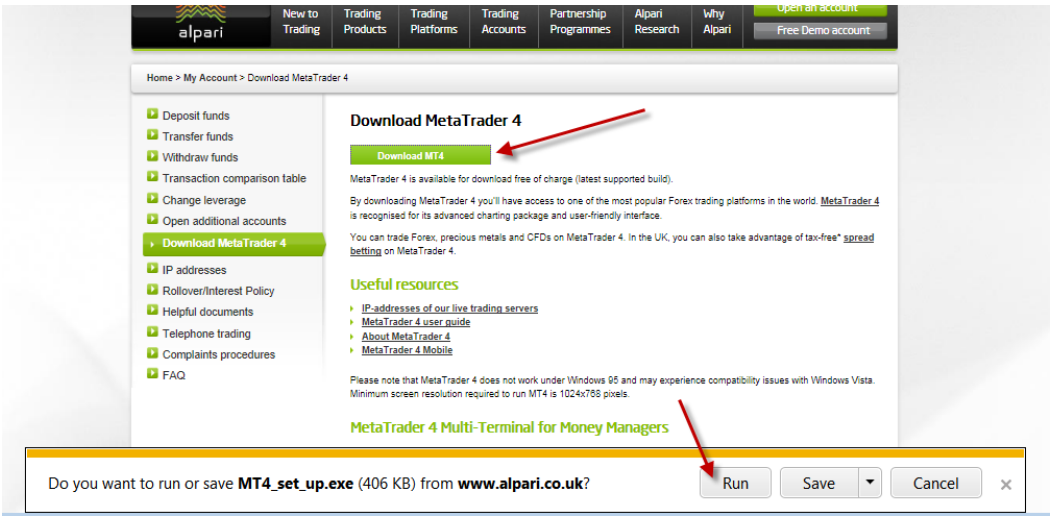
Kind regards,
Alpari (UK)

Step 1: Download and install MetaTrader 4.

Step 2: Test your trading strategies with your Alpari Demo account.

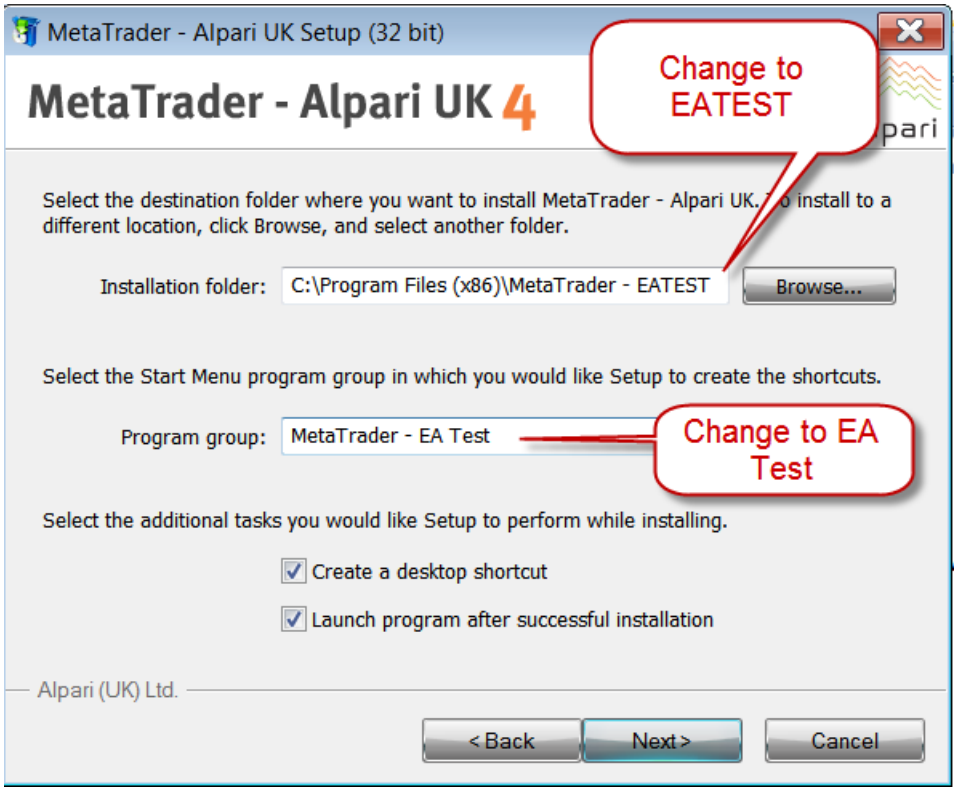
Step 3: Upgrade to a Live account and start trading from just $200!

You will be taken to a download webpage. Click on “Download MT4” and then “Run”



You will be taken to the initial download page. Click “Next>” to continue.

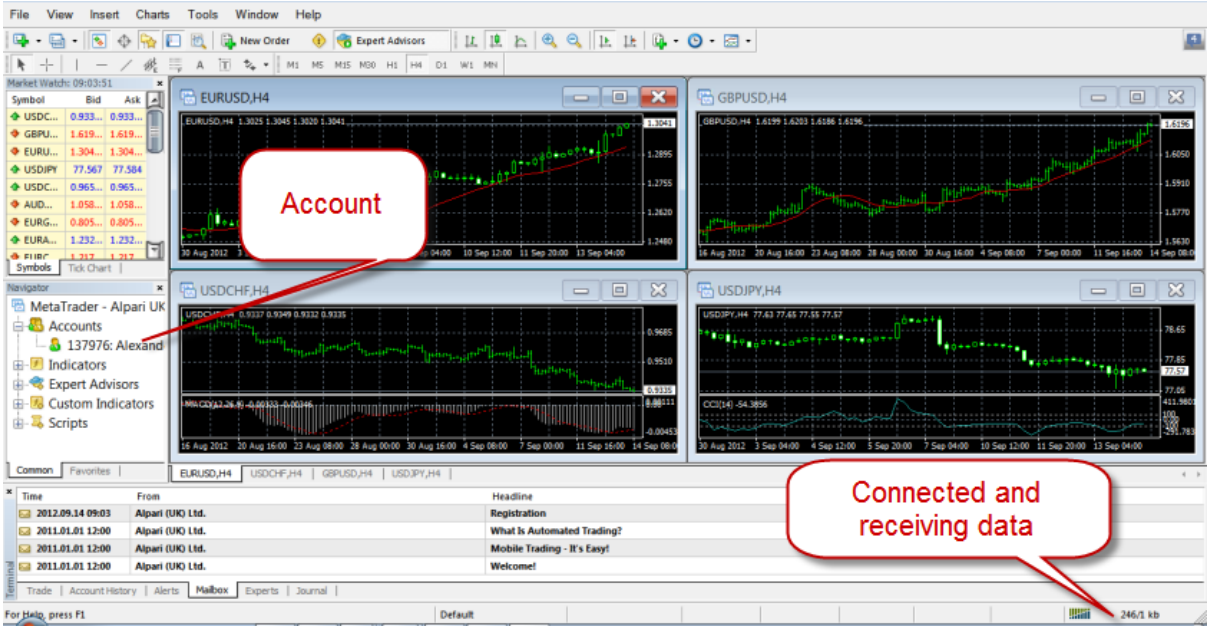
The next section is very important. If you get this wrong the installation will overwrite your current installation (if you have one). You need to change the name of the installation folder and program group as shown below and then click next. If you do not have a previous MT4 installation do this anyway.



When you click “Next” the procedure will automatically install and open your MetaTrader platform. The installation process will place an icon on your desktop and also ask you to open an account. Click Cancel on this pop-up.

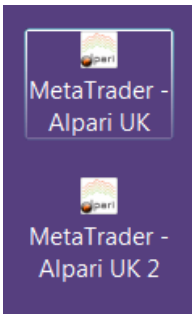
The next pop-up requires the Login and Password. These are contained within the original email you received, so enter them into this window. Make sure you select the appropriate Server for your personal Micro Demo Account as shown below. The actual server will vary depending on the broker used and your location in the world.

Then click “Login”. You should see a platform similar to the one below:

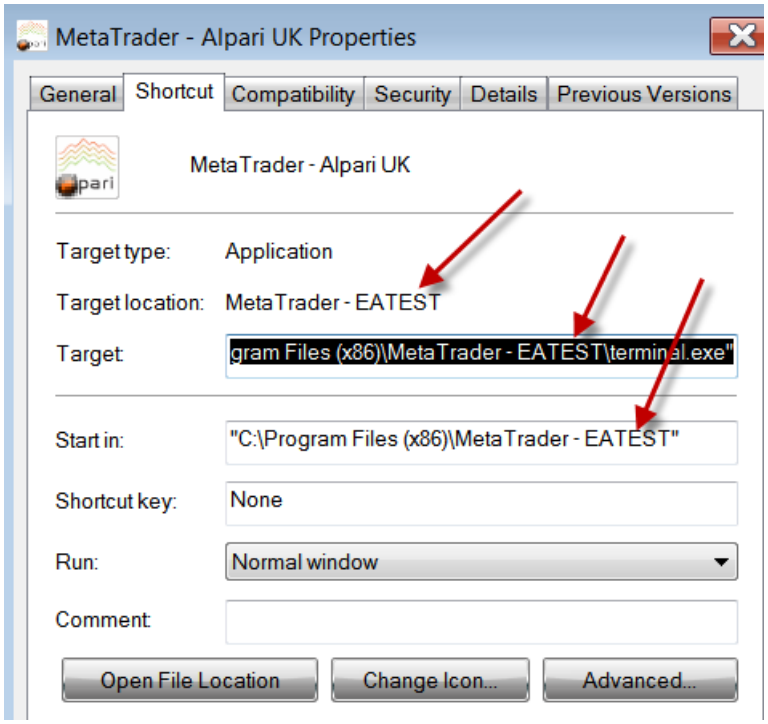


So now we have a test platform for our EA testing and optimisation work.

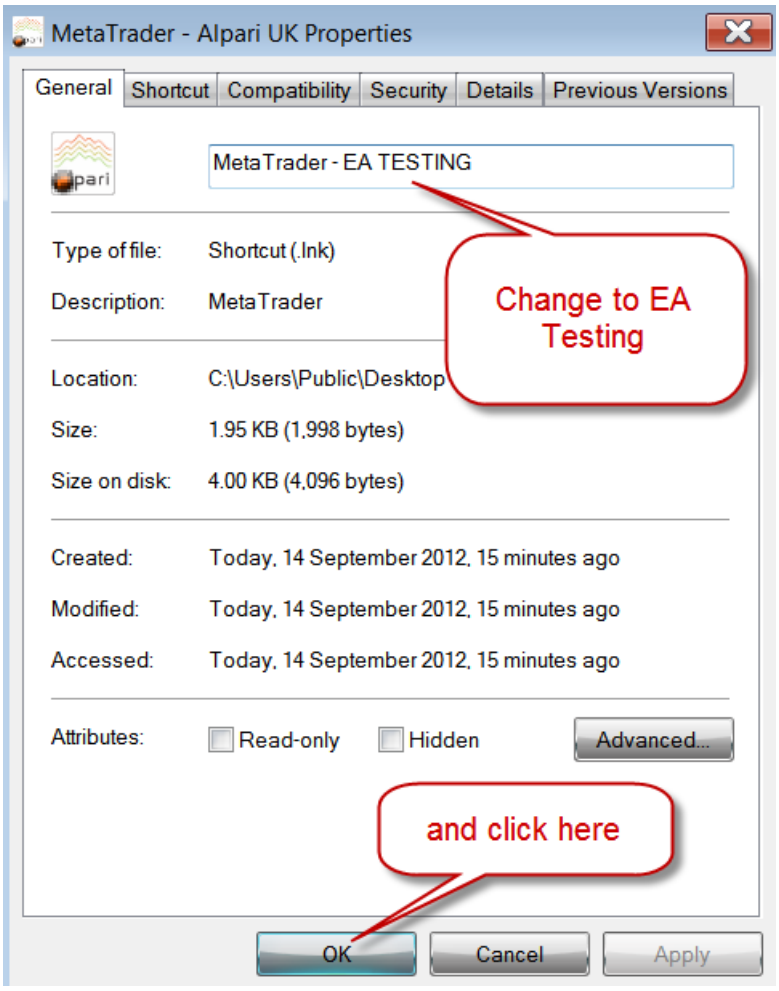
You may now have 2 icons on your desktop with the same names. This occurs if you had previous installations of MT4 before the one just created.



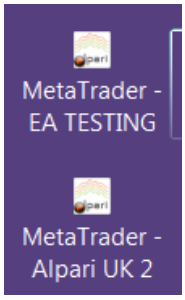
You need to find the test icon. So right click on one of them. Then go to properties at the bottom of the options. Click on properties and look at the directory of the program. If it gives information as shown below then you know that you have found the correct icon.



Then click on the general tab as shown in the picture below and change the icon name as shown and click "OK". You can use any name you like as long as it identifies the EA Testing installation.



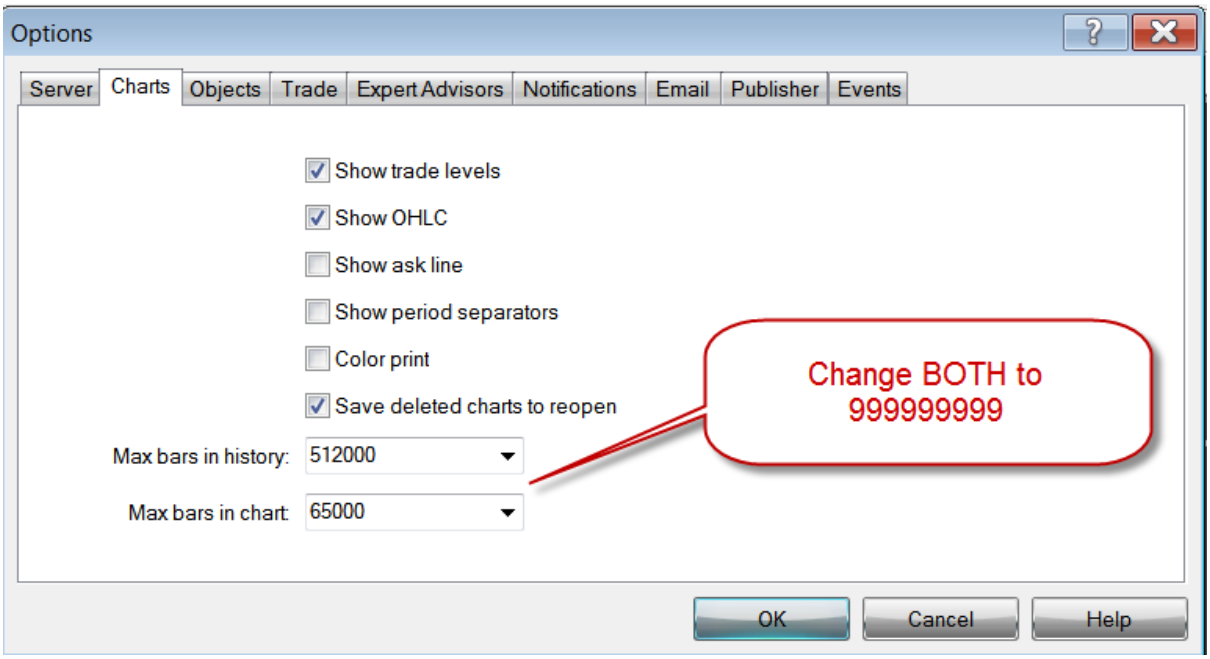
Your icon will have changed and you can now find the test platform much more easily.



OK. Now we are ready to start setting up our data.

2. Settings for the maximum amount of test data

The First step is to go to TOOLS in the main menu. Then select OPTIONS and then charts. You should see the screen below. We want the maximum amount of data for testing and optimisation so change BOTH figures shown to 999999999 and click on OK.

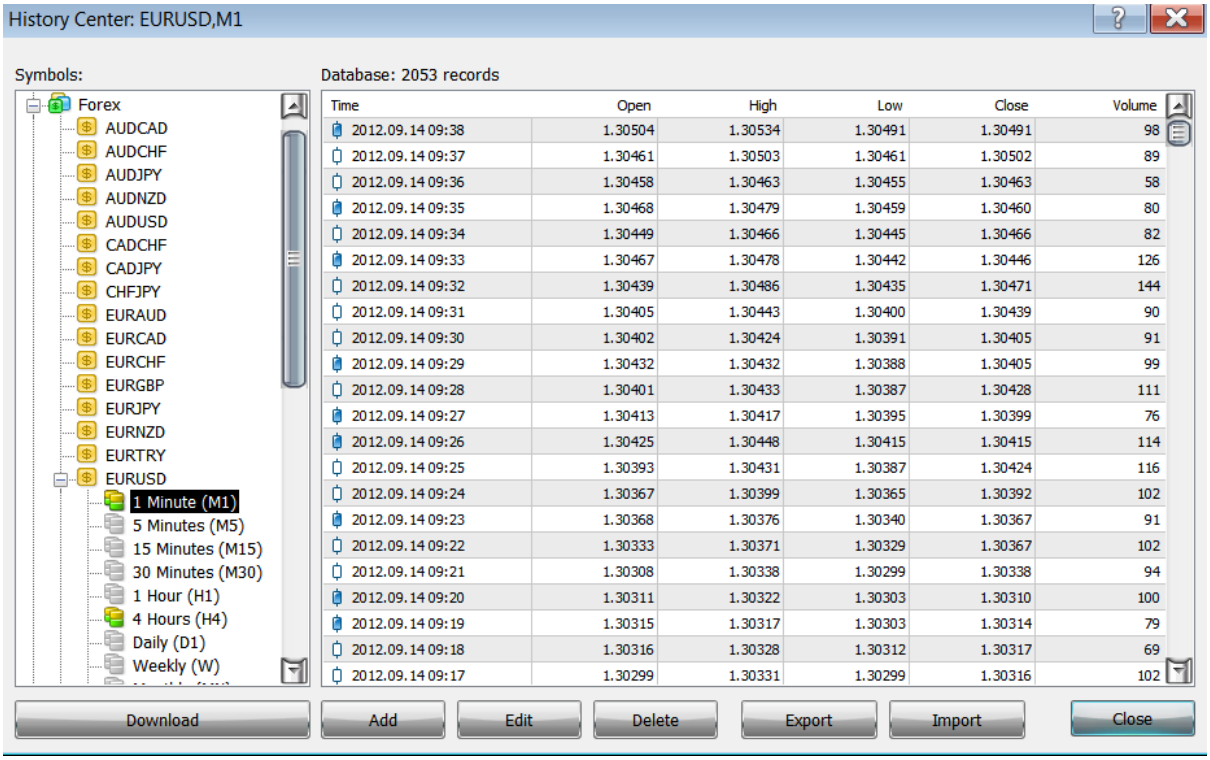


Later on the figures will change to the actual maximums allowed by MetaTrader so don't worry about that.

3 Deleting the old data in your history

The next steps are to delete old data currently on your EA Testing MT4 platform and replace that data with the maximum amount of data and a more complete range of data.

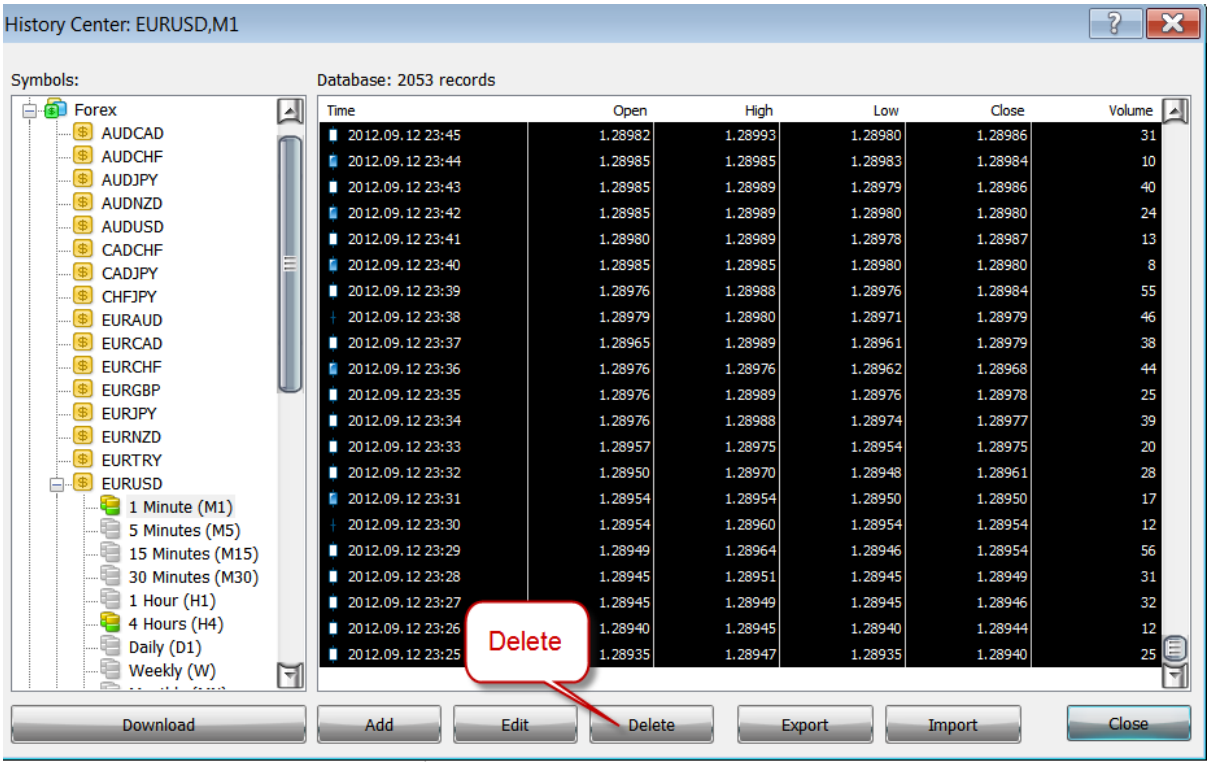
To delete the data click on TOOLS on the main menu. Then “HISTORY CENTER”. Open the Forex tab and you will see a list of currencies available. We will be using the EURUSD in this example. Click on the EURUSD and then double-click on the 1 minute option. You may see nothing or you may see a complete list – as below. If you see nothing move to step 4.



Note that there are only 2053 records

We want to delete this list so

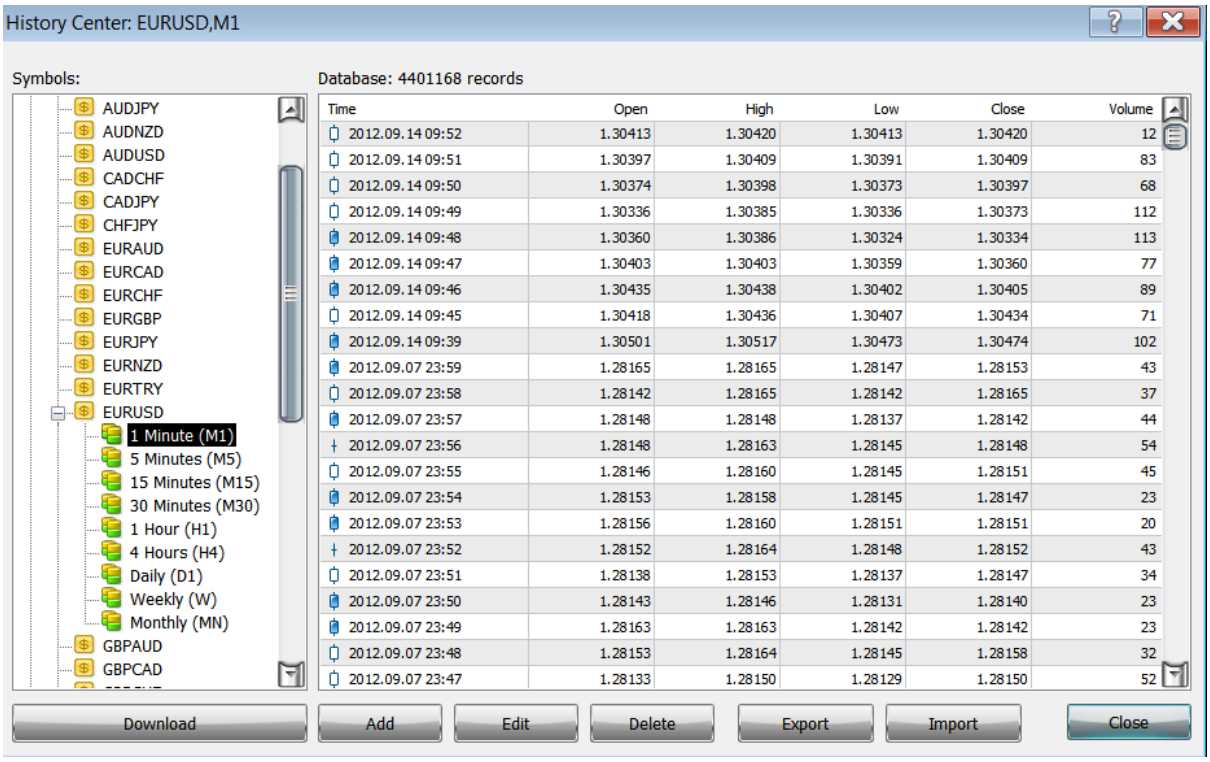
- Highlight the first entry;
- Then scroll down to the last entry using the slide facility on the side;
- Press shift and click on the last entry;
- This should highlight all the entries;
- Then click on delete.



This should delete all the data. Don't worry about the other time frames – they are updated from the 1 minute data.

4 Downloading new test data

Then click on download with the 1 minute highlighted. The download may take some time but you should see something like the screen below when the download is complete.



Double click the other EURUSD timeframes to populate the data to them. They will all become green as shown.

We now have over 4.4 million bits of data for the 1 minute charts. A big improvement of the amount we had before (2053).

<u>You will have to do this for every currency you will be testing</u>. This will ensure that we have all the data we need to do our testing. At this stage the EURUSD data will do. You can come back to this process later to download any other currencies you would like to test.

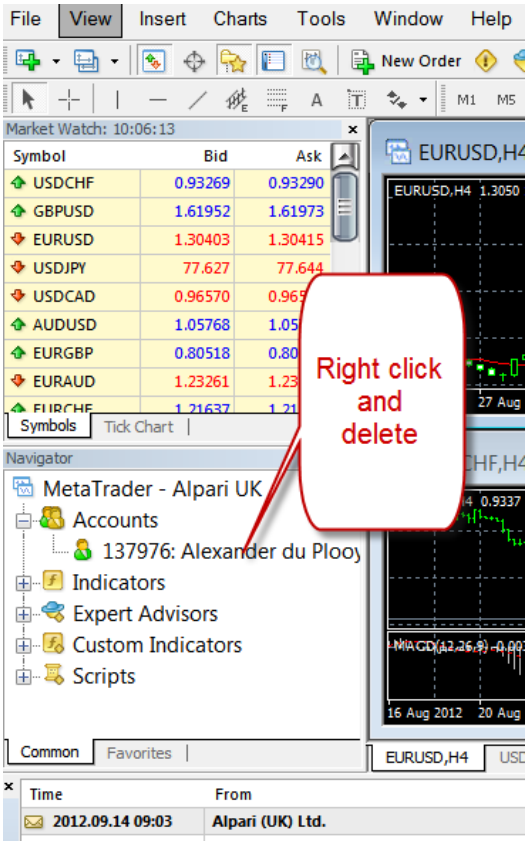
There are other ways of getting much higher quality tick data but because our Magic Moving Average makes decisions based on the close of a candle the MetaQuotes data will be fine for our purposes.

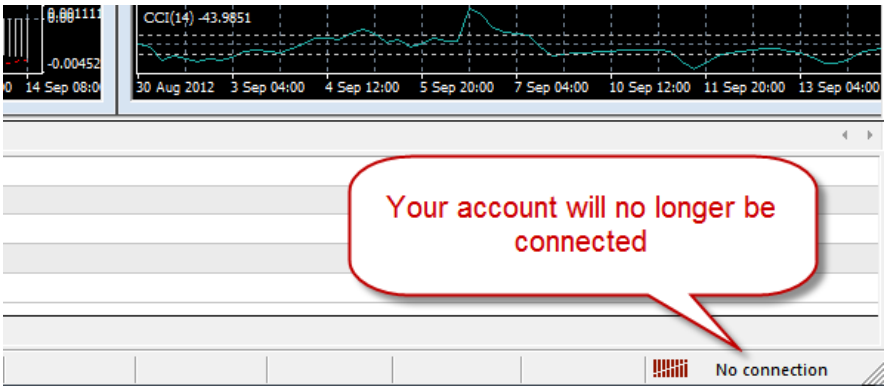
5 Protecting your test data

Now that we have downloaded new test data from metaqoutes.net we don't want to mix our broker data with test data. Already some new 1 minute data will have come in. So to do this we have to delete the account that we opened to turn the test MetaTrader platform into a static offline testing platform.

To do this, open your navigator by going to "View" in the main menu, and selecting "Navigator" if it is not open already.

Find your account in the navigator window. Right click on the account and select delete.





Your platform will no longer be connected and will no longer be updated. **That is what we want as that will keep our test data pure.** When MetaQuotes data is mixed with broker data we will get mismatched data errors when doing our testing. See Guide 3 for more details.

Now close your platform. When you next open you test platform you will be asked to open a new account and to log in. IGNORE these messages by selecting “Cancel” each time. We want to work offline.

We do not need to be connected to the internet to perform the testing and optimisation steps.

6 Updating your test data in the future

As time goes on you will want to update your test data to include the latest history. You may also want to add many more currencies. We suggest that this is done at least every 2 to 3 months if you are an active Expert Advisor user.

This is done by:

- Opening an account by clicking on "File" and "Open an Account". This will re activate your internet connection to enable the downloading of new test data.
- Then repeat steps 3 to 5 in this guide. Your test data will then be updated.

7 Conclusions

We should now have achieved our objectives.

The objectives of Guide 2 are to help you:-

- Create a separate test facility for testing all your Expert Advisors
- Download a considerable amount of historical test data

The optimisation and testing of the Magic Moving Average Expert Advisor will be covered in separate guides. If you have completed the above steps you are ready to move to the fun part of using this Technique which we will cover the Optimisation and Testing guides next.

8 Questions and Answers

If you have any questions or comments to make about the content of the Guide please send an email to info@expert4x.com with the heading "Magic MA Guide 2 – Question and Comments" .

Below are a number of questions and comments already received:

Question 1: NO NEW DATA message

My broker is FX Primus and I've installed a new MT4 especially for the EA Test as suggested in the Guide. However, I'm having problems downloading the history data after I deleted the original history. I get a "There are no new data for the symbol EURUSD-".

Answer 1: NO NEW DATA message

If you compare your History Centre data with the data in the course you will see that your currencies are described with a dash behind them EURUSD- . The course data is just EURUSD . The dash is preventing to download of MetaQuotes data as MetaQuotes only has data for EURUSD not EURUSD- . It therefore does not look like you can use your brokers demo account and will have to use an Alpari UK demo account.

There is a technical solution to your problem which is complicated but may help – This solution was created by one of our tester who used it for Forex.com when they had the same problem as you are experiencing. Their currencies had pro added (EURUSDpro) compared to your dash. Please note that we have no experience with this solution so cannot support it. You can download this solution from the link below.

IMPORTING TEST DATA

So your options are to either use an Alpari UK demo account or import data into your demo account using the 2nd approach.

Question 2: Free and Paid VPS Services.

I've been testing and optimising with great results, I did read that with this would be 3 months access to a VPS service, I cannot seem to find any info on it. Can you help?

Answer 2: Free and Paid VPS Services.

Please follow this link to the 3 month free VPS broker offer. http://expert4x.com/free-forex/expert4x-forex-products-and-services-for-free-from-avatrade/

We use 4XSolutions – please follow this link for more details.

http://expert4x.com/associated-forex-services/forex-vps-service-provider/

Question 3: Historical Data upload alternative.

I've followed the manual instructions to download M1 data, but had to delete it as there was a big shift of 66 pips. Now when I try to download MQ data it tells me there is no new data to download. All I can see is data for 2012, no earlier. But, when I go to open files offline, I can see the EURUSD TF has over 430,000 bars of data but is only showing. All other TFs are showing data back to 1999.

Answer 3: Historical Data upload alternative.

Technical Support: I experienced similar issues a few times and when this issue becomes permanent I actually go for out-of-box solution: instead of deleting the historical data within MT4 application, I quit the application and delete the historical data externally. If you ready for that follow these steps:

1. Close off all chart windows in the running MT4;
2. Quit MT4;
3. Delete unwanted history data, i.e. search for *.hst files within MT4 installation like C:\Metatrader path\history\demo_account folder;
4. Startup MT4;
5. Follow the normal data download steps as usual.

The step 1-4 should clean up the historical data repository. When step 4 starts up MT4 again do not create new chart window as that triggers real-time download of recent data, instead go straight for History Centre to download off-line data.

Question 4: Broker data differences from MetaQuotes

Why do you suggest that we test using data from MetaQuotes.net and not the broker that we intend to use? I am planning on using Alpari UK. Have you found consistency between MetaQuotes and Alpari and am I worrying unnecessarily?

Answer 4: Broker data differences from MetaQuotes

You may have misread the warning in the Guide. It actually says that you must make sure that your broker data is the same as the MetaQuotes data by giving you instructions on how to do this comparison.

It seems logical that it is **absolutely critical** that the test data you use represents the actual data you are going to use when trading live.

It is up to you as the trader to determine this for yourself. We can give no assurances in this area at all.

If there are meaningful differences you will have to use your broker's data for testing.

Question 5: Test data dates

I have down loaded test data from MetaQuotes using FinFx(my live broker) on Demo following the steps in Guide 2. I noticed that the data shows me 4597156 records and ends with the date (2012.01.13 14:06). I found the dates to be odd being that it is already Oct 6th 2012. Is there something that I am doing incorrectly? Please advice. Thank you for your time.

Answer 5: Test data dates

As mentioned in the Guide if the quality of your test data is not good using your own broker platform consider opening an Alpari UK demo account for testing purposes.

Question 6: Making sure Broker and test data agree

How do I know that my test data agrees with my broker data I am going to use for live trading?

Answer 6: Making sure Broker and test data agree

Please do not test using data that is different from your live data. Time differences are OK but differences in volumes and price levels are not. If is very easy to test for this. Go to your Live trading MT4 platform and click on "Tools" then "History Center" than on select a currency and timeframe that you will be using. Then select a time and date about 3 months back and note the values for that time and date.

2012.04.17 00:30	1.31343	1.31358	1.31326	1.31345	331
2012.04.17 00:15	1.31364	1.31379	1.31333	1.31335	199
2012.04.17 00:00	1.31411	1.31412	1.31324	1.31361	299
2012.04.16 23:45	1.31393	1.31424	1.31389	1.31412	485
2012.04.16 23:30	1.31368	1.31397	1.31366	1.31389	275

Then do the same for your test data. Compare the values – they should be the same. It not check to see if there are broker time differences and adjust for that a recheck – the values should be the same.

2012.04.17 01:00	1.31313	1.31348	1.31293	1.31309	369
2012.04.17 00:45	1.31344	1.31398	1.31295	1.31315	362
2012.04.17 00:30	1.31343	1.31358	1.31326	1.31345	331
2012.04.17 00:15	1.31364	1.31379	1.31333	1.31335	199
2012.04.17 00:00	1.31411	1.31412	1.31324	1.31361	299
2012.04.16 23:45	1.31393	1.31424	1.31389	1.31412	485
2012.04.16 23:30	1.31368	1.31397	1.31366	1.31389	275
2012.04.16 23:15	1.31352	1.31372	1.31349	1.31367	374

If they differ you need to consider changing your broker to one that aligns with the test data that you will be using or use you broker data for testing.

Question 7: Using Broker test data only.

Can I get by without checking the data variances with my demo vs. live account? I will be using HotForex. The rage of profit seems great enough, along with the research I've done, to not worry about any minor price fluctuations. Just how important is the difference? Is it possible to skip this step?

Question 7: Using Broker test data only.

Yes it is possible to skip this step and use your own Brokers data for testing. The advantage is that you will know that the test data is fully aligned to your broker data.

The disadvantage is that the broker data is sometime incomplete and is almost always highly restricted in terms of quantity. Often you can only go back 3 weeks on the 1 minute charts.

You can overcome this by changing your Tools / Options / Chart maximum bars in history and on the charts to 9999999999 as shown on page 11 of Guide 2.

You would then scroll to the end of the chart you want to test. This is the left-hand part the chart where it has no more history to show. If you then try to force scroll further by using the back arrow or the mouse you may find the charts updating automatically the more you scroll.

You may also find that when you use the Strategy Tester and specify a date further back from the data available to you in history that the data is automatically updated by this action when you run the test.

By the way you can see how much test data you have by clicking on "File" and then "Open offline" in the MetaTrader menu.

Hope this helps with your decision to use broker or MetaQuotes data.